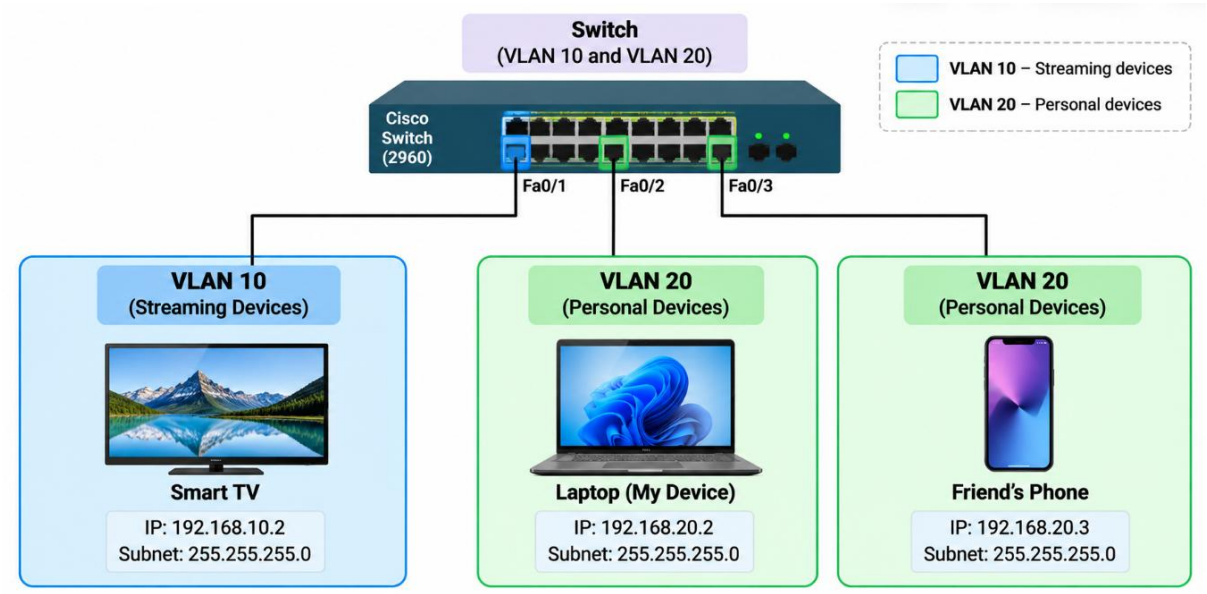
VLAN Configuration

1. Network Diagram with VLANs

A VLAN (Virtual Local Area Network) divides one physical switch into separate logical networks. Devices in different VLANs are isolated at Layer 2 unless a router or Layer 3 switch is configured to route traffic between them.



VLAN 10 and VLAN 20 are separate broadcast domains.

Example IP plan:

VLAN	Purpose	Example Network
VLAN 10	Streaming devices	192.168.10.0/24
VLAN 20	Personal devices	192.168.20.0/24

A switch can separate these VLANs, but devices in different VLANs need a router or Layer 3 switch to communicate with each other.

2. Create Two VLANs in Cisco Packet Tracer

Step 1: Add devices

In Packet Tracer, place one Cisco 2960 switch and five PCs. Connect each PC to the switch using Copper Straight-Through cables.

Suggested port assignments:

PC	Switch Port	VLAN	IP Address
PC0	Fa0/1	10	192.168.10.2
PC1	Fa0/2	10	192.168.10.3
PC2	Fa0/3	10	192.168.10.4

PC3	Fa0/4	20	192.168.20.2
PC4	Fa0/5	20	192.168.20.3

Use subnet mask 255.255.255.0 for all five PCs. No default gateway is required for testing connectivity within each VLAN.

Step 2: Create VLANs

Click the switch, open the CLI, and enter:

enable

configure terminal

vlan 10

name STREAMING

exit

vlan 20

name PERSONAL

exit

Step 3: Assign ports to VLAN 10

interface range fastEthernet 0/1-3

switchport mode access

switchport access vlan 10

exit

Step 4: Assign ports to VLAN 20

interface range fastEthernet 0/4-5

switchport mode access

switchport access vlan 20

exit

end

Step 5: Verify VLAN configuration

show vlan brief

You should see Fa0/1–Fa0/3 assigned to VLAN 10 and Fa0/4–Fa0/5 assigned to VLAN 20.

Step 6: Test connectivity

Open each PC's Desktop → Command Prompt.

- From PC0, run ping 192.168.10.3 — this should succeed because both PCs are in VLAN 10.
- From PC0, run ping 192.168.20.2 — this should fail because PC0 and PC3 belong to different VLANs, and no inter-VLAN routing is configured.

Conclusion: PCs within the same VLAN can communicate, while PCs in different VLANs cannot communicate without a Layer 3 routing configuration.

3. Move a PC from VLAN 10 to VLAN 20

Suppose PC2 is connected to switch port Fa0/3. Initially, it belongs to VLAN 10. We will move it to VLAN 20.

Before: PC2 belongs to VLAN 10

enable

configure terminal

interface fastEthernet 0/3

switchport mode access

switchport access vlan 10

end

Set PC2's IP address to 192.168.10.4 with subnet mask 255.255.255.0.

Test with another VLAN 10 PC:

ping 192.168.10.2

Expected result: Ping succeeds.

After: Move PC2 to VLAN 20

On the switch CLI, enter:

enable

configure terminal

interface fastEthernet 0/3

switchport access vlan 20

end

Now change PC2's IP address to 192.168.20.4, keeping the subnet mask as 255.255.255.0.

Verify the change

show vlan brief

Fa0/3 should now appear under VLAN 20.

Test from PC2:

ping 192.168.20.2

Expected result: Ping succeeds if the other PC is correctly configured and connected.

Then try:

ping 192.168.10.2

Expected result: Ping fails because PC2 is now in VLAN 20 and PC0 remains in VLAN 10, with no inter-VLAN routing configured.

Conclusion: Changing the access VLAN on the switch port moves the connected PC into the new VLAN. The PC's IP address must also match the new subnet.

4. Automate Assigning 10 Switch Ports to VLAN 30

Cisco IOS supports the interface range command, which lets us configure multiple ports together without repeating the commands.

Configuration commands

enable

configure terminal

vlan 30

name VLAN30

exit

interface range fastEthernet 0/1-10

switchport mode access

switchport access vlan 30

exit

end

Explanation

Command	**Purpose**
vlan 30	Creates VLAN 30.
name VLAN30	Assigns a name to the VLAN.
interface range fastEthernet 0/1-10	Selects switch ports Fa0/1 through Fa0/10 together.

switchport mode access	Configures the ports as access ports.
switchport access vlan 30	Assigns all selected ports to VLAN 30.
end	Exits configuration mode.

Verify the configuration

Run:

show vlan brief

Expected result: VLAN 30 should list ports Fa0/1 through Fa0/10.

Note: This configuration assumes a switch model with those FastEthernet ports, such as a typical Cisco 2960 in Packet Tracer. If your switch uses GigabitEthernet ports or a different numbering scheme, adjust the interface range accordingly.